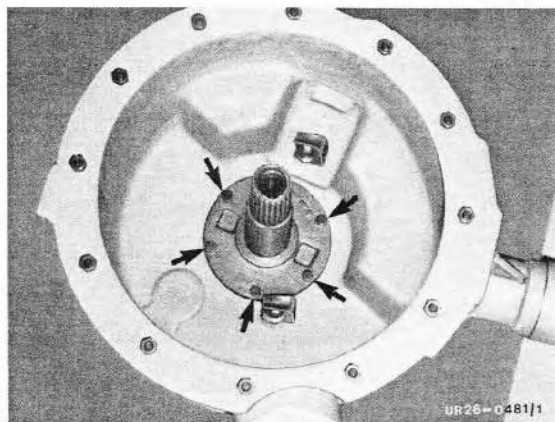




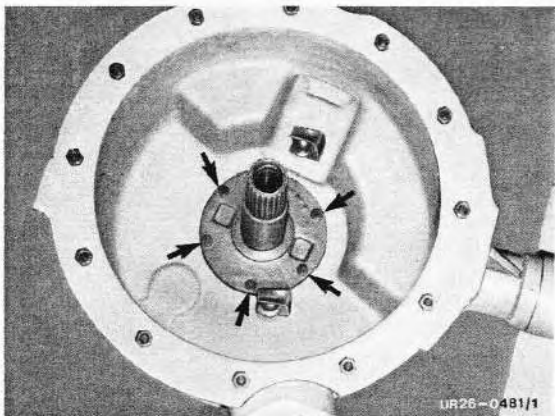
25 Unscrew fastening bolts, remove guide flange and detach shaft seal.

26 Clean and check all parts, exchanging if necessary.



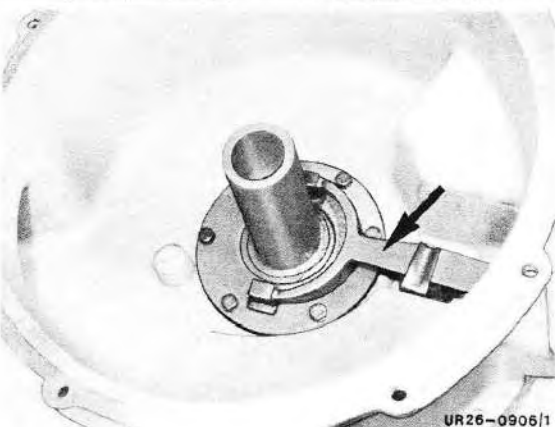
Assembly

1 Coat shaft seal with lubricating grease No. 2 and force into position using fitter No. 8.



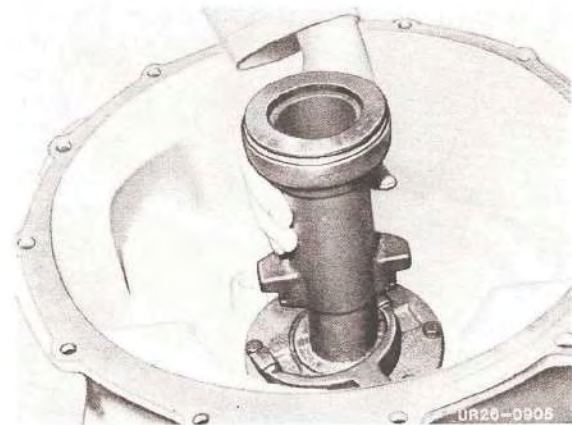
2 Coat guide flange with sealant No. 1 and then insert.
See 1.4/1 for tightening torques.

Note: Watch out for oil passage.

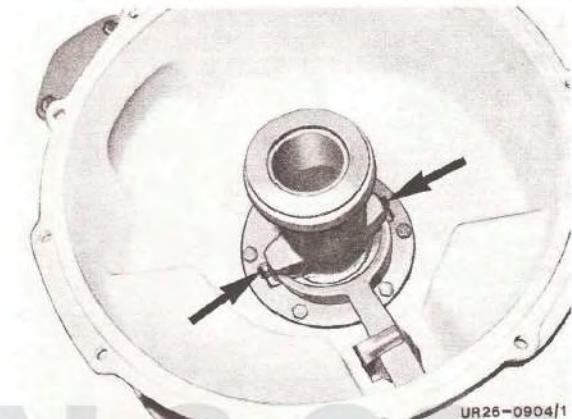


3 Fit main clutch throwout lever.

- 4 Fit throwout bearing.

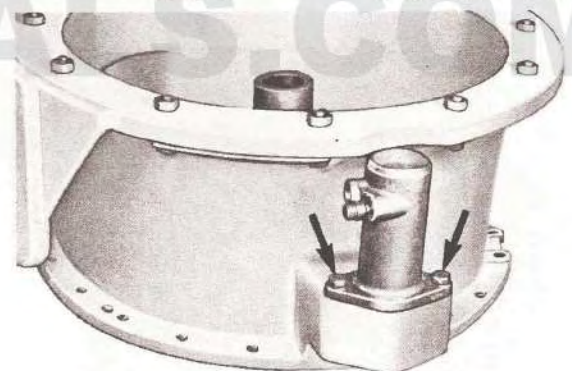


- 5 Attach springs to throwout lever.

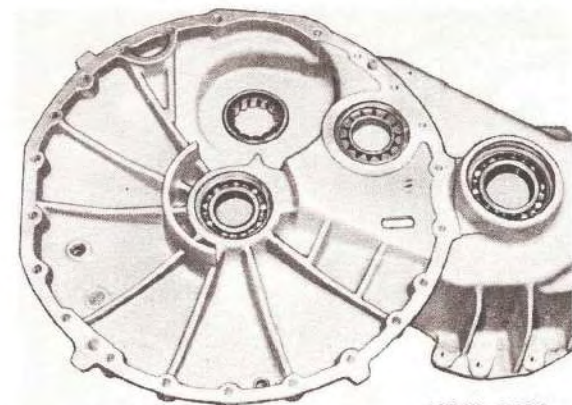


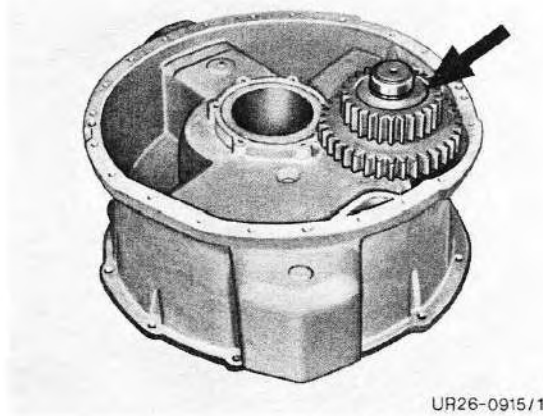
- 6 If necessary, coat clutch slave cylinder with sealant No. 4 and bolts with locking fluid No. 5 prior to attaching.
See 1.4/1 for tightening torques.

- 7 Turn clutch casing by 180°.

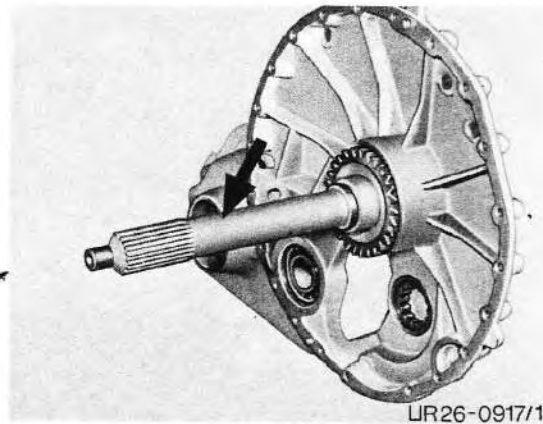


- 8 Introduce roller bearing into transmission housing and secure with snap ring.

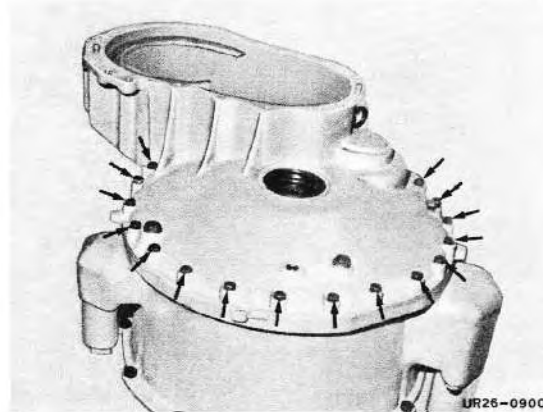




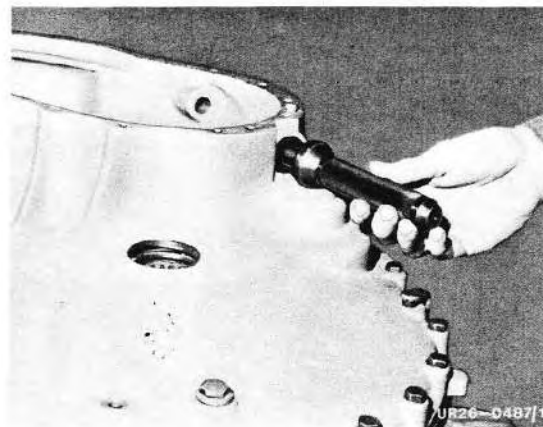
9 Install idler gears.



10 Press transmission input shaft into position.

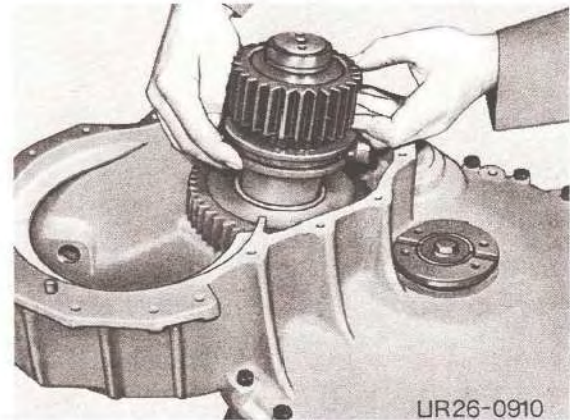


11 Attach and tighten transmission housing with gasket, referring to 1.4/1 for tightening torque.

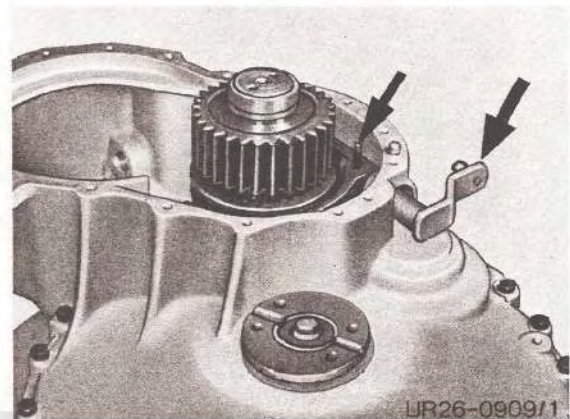


12 Coat shaft seal with lubricating grease No. 2 and force into position using special tool No. 7.

- 13 Install input shaft with selector fork.
See 1.4/1 for assembly of input shaft.

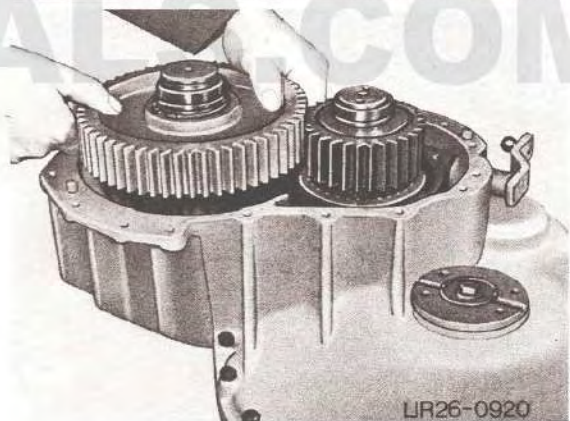


- 14 Install and secure shift shaft.

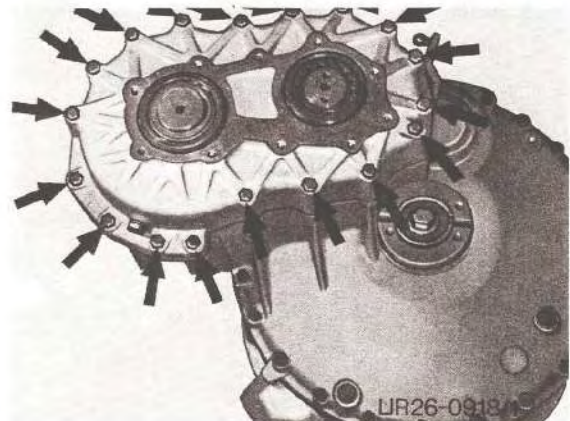


Valid for power take-off at front

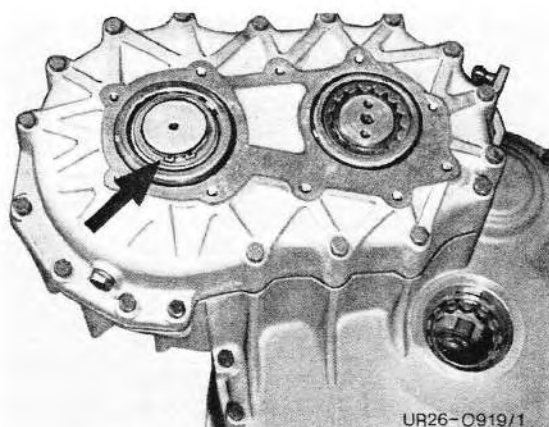
- 15 Install output shaft.



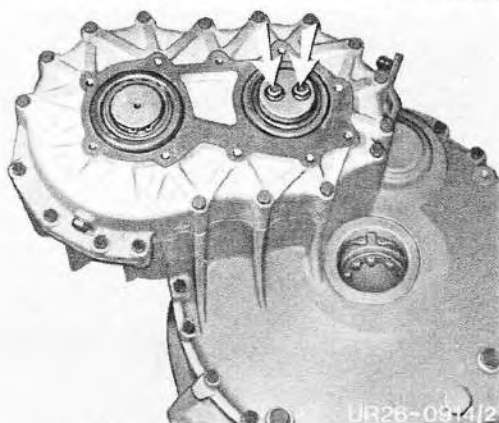
- 16 Force outer races of roller bearings into transmission housing, attaching and tightening roller bearings with gasket.
See 1.4/1 for tightening torques.



SA 35 737



17 Fit thrust washer and snap ring.



18 Attach thrust washer and supporting disk. Tighten supporting disk with reference to 1.4/1 for tightening torques.

Note: Self-locking bolts are to be used once only.



19 Attach and tighten end cover with gasket. See 1.4/1 for tightening torques.

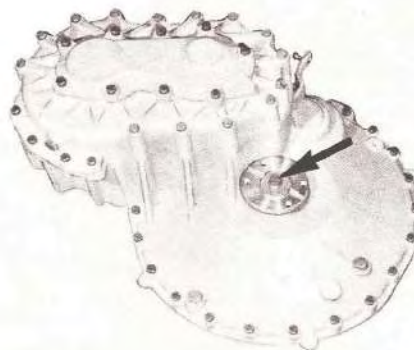


20 Coat both shaft seals with lubricating grease No. 2 and force into position using special tool No. 2.

21 Coat input shaft splining with paste No. 2, attach flange and tighten supporting plate with sealing ring.

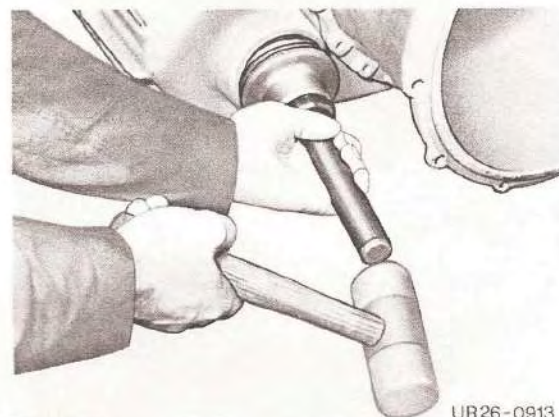
Note: Self-locking bolts are to be used on e only.

22 Turn PTO transmission by 90°.



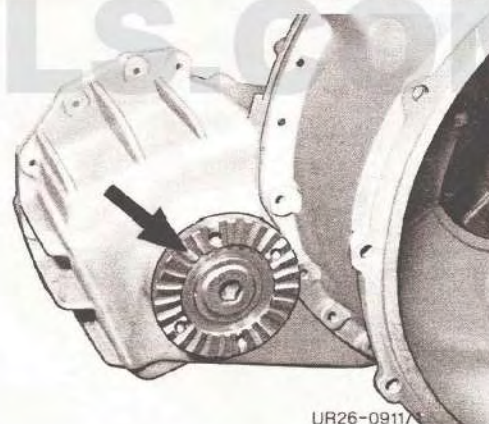
UR26-0932/1

23 Coat both shaft seals with lubricating grease No. 2 and force into position using special tool No. 3.



UR26-0913

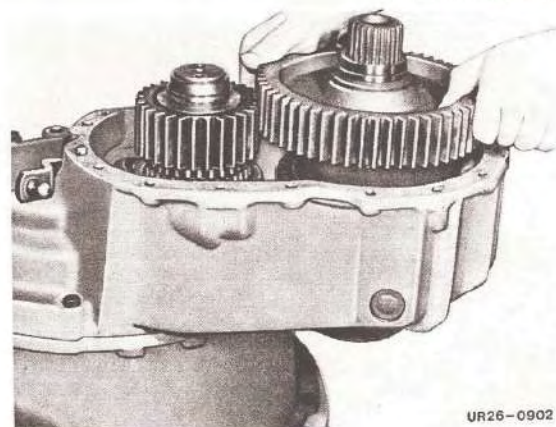
24 Coat output shaft spline with paste No. 3. Attach flange and tighten screw plug with sealing ring. Secure screw plug with aid of center punch. See 1.4/1 for tightening torques.



UR26-0911/

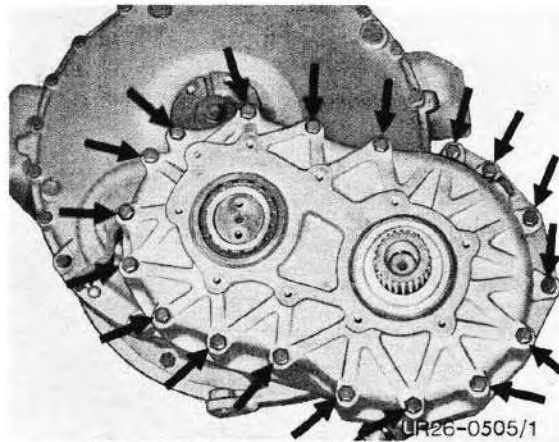
Valid for power take-off at front and rear

25 Install output shaft.

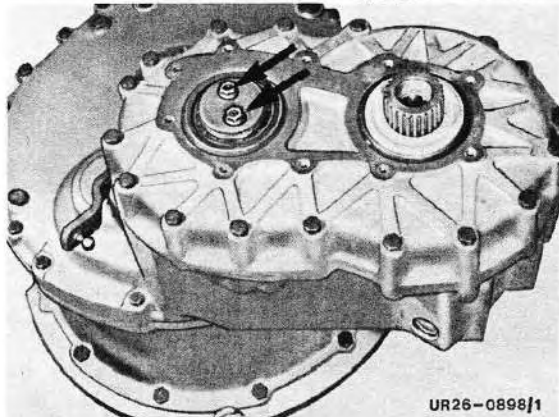


UR26-0902

SA 35 737

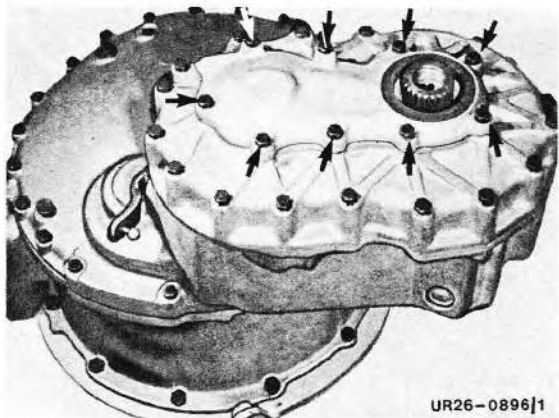


26 Force outer races of roller bearings into transmission housing, attaching and tightening with roller bearings and gasket.
See 1.4/1 for tightening torques.

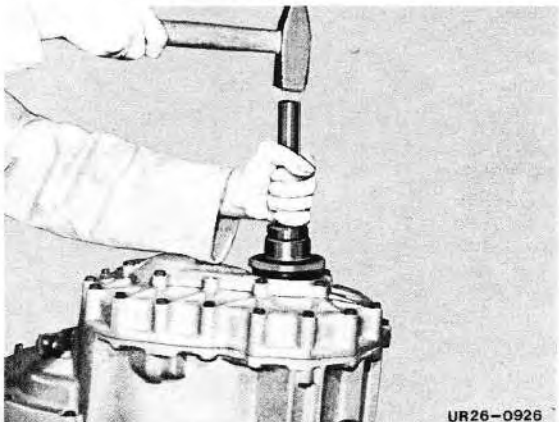


27 Attach thrust washer and supporting disk. Tighten supporting disk with reference to 1.4/1 for tightening torques.

Note: Self-locking bolts are to be used once only.

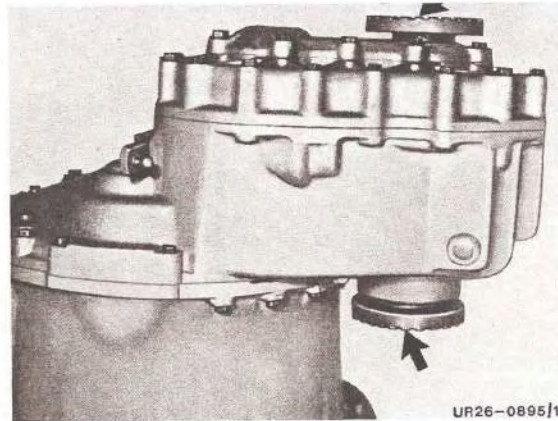


28 Insert Nilos ring, attaching and tightening end cover with gasket.
See 1.4/1 for tightening torques.



29 Coat shaft seal with lubricating grease No. 2 and force into position using special tool No. 4.

- 30 Coat both output shaft splines with paste No.3, attach flange, insert and secure screw plug.
See 1.4/1 for tightening torque.



- 31 Coat shaft seal with lubricating grease No. 2 and force into position using special tool No. 2.



- 32 Coat input shaft spline with paste No. 3, attach flange and tighten supporting plate.

Note: Self-locking bolts are to be used once only.



Contents

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Consumables	1.5/1
Tightening torques	1.5/1
Exploded view	1.6/1
2 Removing and installing fast PTO	2.1/1
3 Disassembly and assembly of fast PTO	3.1/1
4 Disassembly and assembly of shift housing	4.1/1

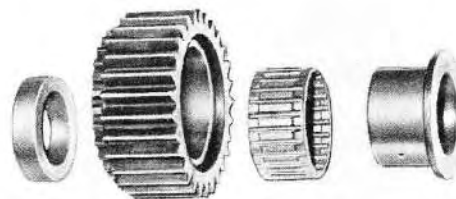
Installation Survey

Chassis Model		Main transmission Model		PTO Ratio	Direction of rotation	Installation
	Sales Designation		Sales Designation			
435.110	U 1700	717.901	UG 3/40-8/13,01 GPA	1	left	SA 35 925
.111	U 1700 L					
.115	U 1300 L					
.117	U 1300 L/37					
.160	Attachment TM 170	717.931				Series
.170	Attachment Condor	717.930				

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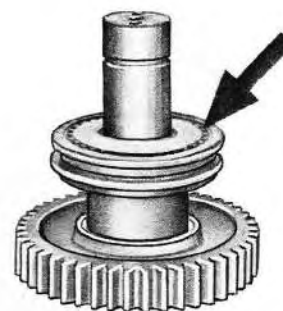
Disassembly

- 1 Remove drive gear 540/min with roller bearing. Detach bearing bushing, needle bearing and spacing ring.



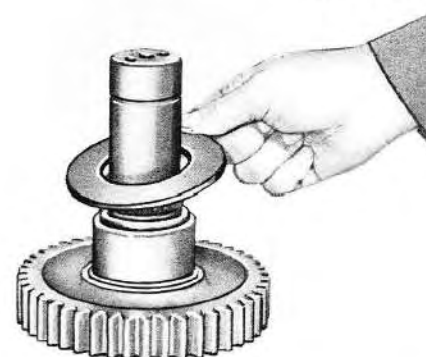
UR26-0493

- 2 Remove sliding shift collar with dog.



UR26-0907/1

- 3 Detach thrust washer and spacing sleeve.
- 4 Clean and check all parts, exchanging if necessary.

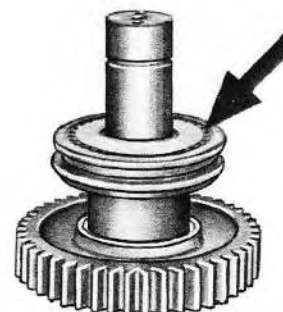


UR26-0916

Assembly

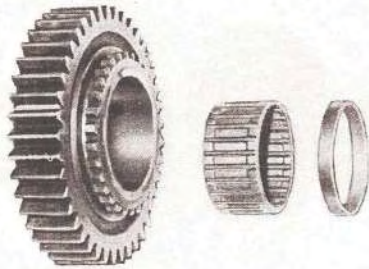
- 1 Fit spacing sleeve and thrust washer.
- 2 Insert sliding shift collar with springs and balls, slipping both parts onto input shaft.

Note: Watch out for balls and springs.



UR26-0907/1

SA 35 737



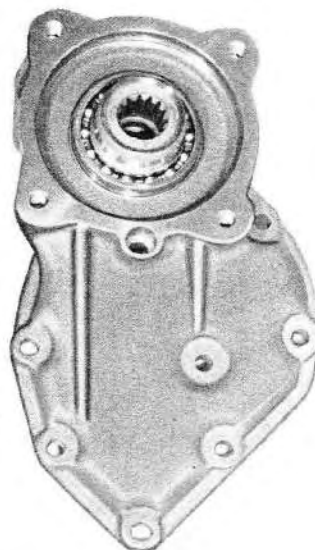
UR26-0495

3 Insert spacing ring, drive gear 540/min, needle bearing and bearing bushing.

4 Attach spacing ring and roller bearing.

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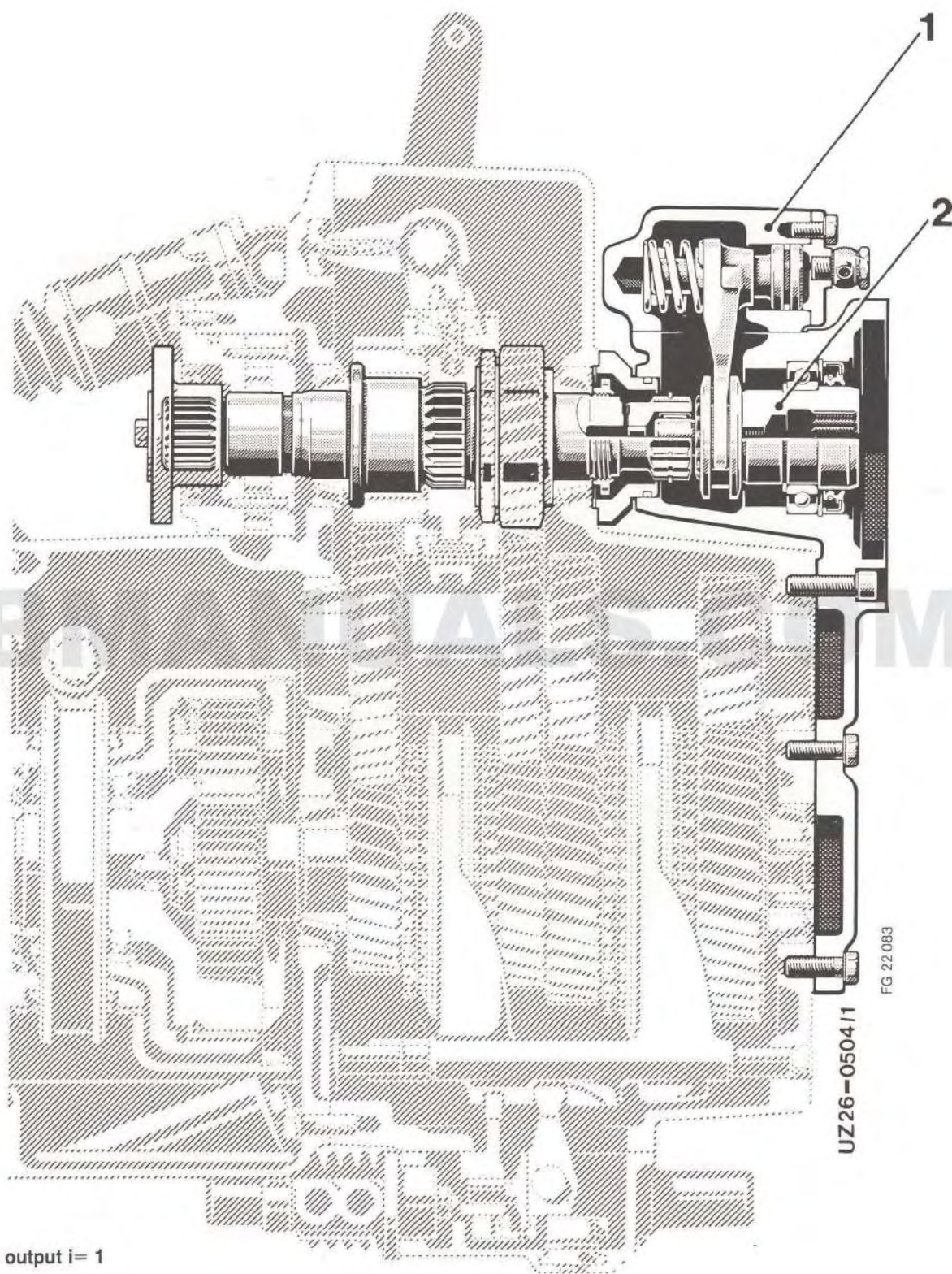
General view



UR 26-1174

PTO

Sectional View

PTO output $i=1$

- 1 PTO output housing
- 2 Implement flange

Technical data**PTO gearing**

Output shaft		
DB Part No.	Output gearing	Gearing designation
425 264 03 06	Splined hub profile	N 25 x 1.25 x 18 x 9 H DIN 5480
425 264 04 06		N 30 x 2.0 x 14 x 9 H DIN 5480
425 264 05 06		N 45 x 3.0 x 14 x 9 H DIN 5480

Quantities

Designation	Service product	SAE grade	Capacity l	Installation
Main transmission	Transmission fluid ¹⁾	80 ²⁾	10.5	SA 35 925
— with PTO		80 W/85 W ²⁾ 85 W — 90 ³⁾ 90 ³⁾		

1) For U 435.111/.113/.115 special vehicles engine oil SAE 10 W

2) Optional

3) In hot regions

Settings

End play of main shaft	mm	0.03 ±0.02
------------------------	----	------------

Consumables

Ser. No.	Designation	Parts No.
1	Loctite No. 573 ¹⁾	001 989 45 20
2	Loctite No. 241 ²⁾	002 989 70 71
3	Long life grease	commercial quality

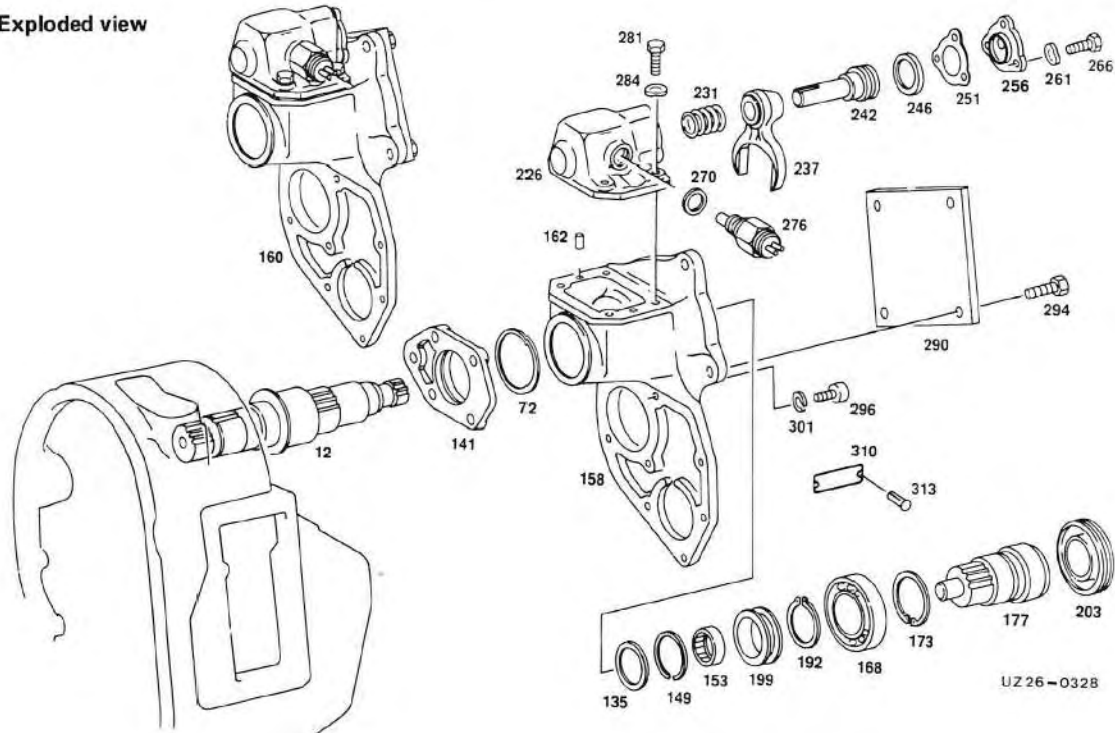
1) Surface sealing compound

2) Screw sealing compound

Tightening torques

Designation	Thread/Strength	Nm
PTO at main transmission	M 10 x 35 — 8.8	33
Shift housing at PTO	M 8 x 30 — 8.8	25
Bearing cover at shift housing	M 8 x 25 — 8.8	25

Exploded view



PTO

- | | | | |
|-----|--------------------------|-----|---------------|
| 12 | Drive shaft | 246 | Sealing ring |
| 72 | Shim | 251 | Gasket |
| 135 | Sealing ring | 256 | Bearing cover |
| 141 | Bearing cover | 261 | Spring washer |
| 149 | Circlip | 266 | Bolt |
| 153 | Needle bearing | 270 | Sealing ring |
| 158 | Housing | 276 | Switch |
| 162 | Straight pin | 281 | Bolt |
| 168 | Deep-groove ball bearing | 284 | Spring washer |
| 173 | Circlip | 290 | Cover |
| 177 | Drive shaft | 294 | Bolt |
| 192 | Circlip | 296 | Bolt |
| 199 | Shift slide sleeve | 301 | Spring washer |
| 203 | Sealing ring | 310 | Unit plate |
| 226 | Shift housing | 313 | Grooved pin |
| 231 | Compression spring | | |
| 237 | Shift fork | | |
| 242 | Shift piston | | |

Removal

- 1 Remove hydraulic pump.
- 2 Unscrew pneumatic supply line and cable for indicator.

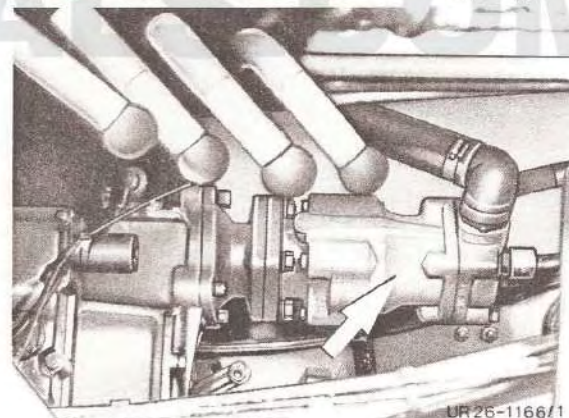
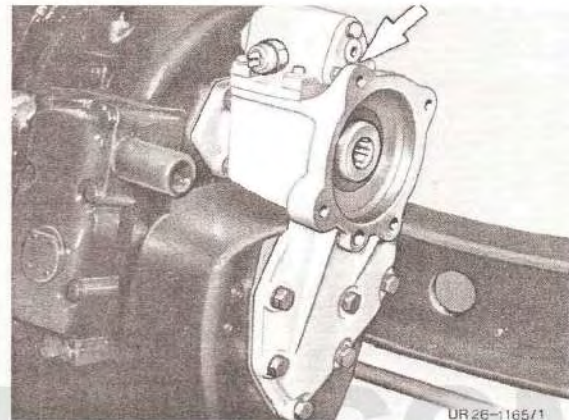
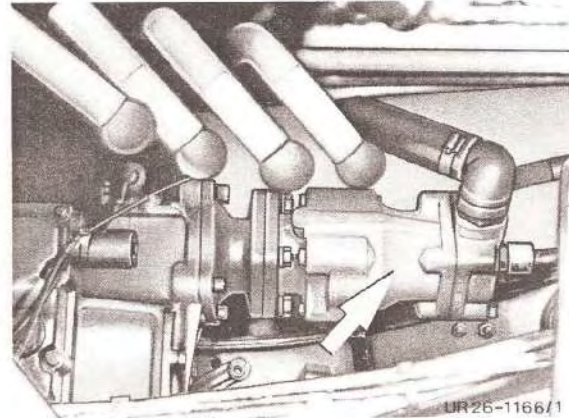
- 3 Remove PTO.

Installation

- 1 Coat contact surface with sealing compound No. 1 and fit PTO.

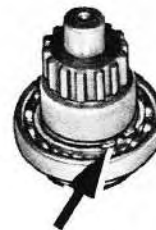
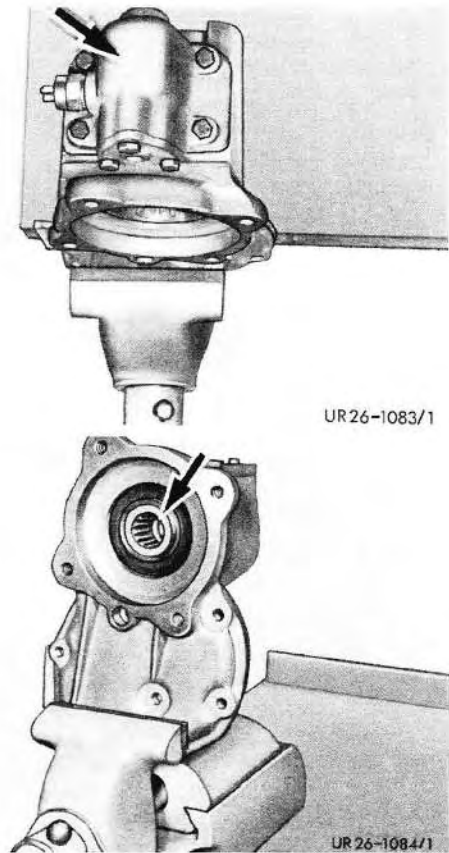
- 2 Screw on pneumatic supply line and cable for indicator.

- 3 Fit hydraulic pump.



Disassembly

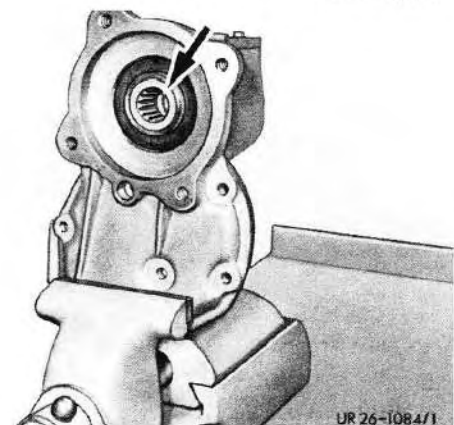
- 1 Remove fast PTO, refer to 2.1/1.
- 2 Remove electric switch and shift housing.
- 3 Disassemble shift housing, refer to 4.1/1.
- 4 Remove shaft sealing ring, release PTO shaft and remove.
- 5 Release deep-groove ball bearing and remove.
- 6 Clean and check all parts, exchange as necessary.
- 7 Disassemble shaft housing, refer to 4.1/1.

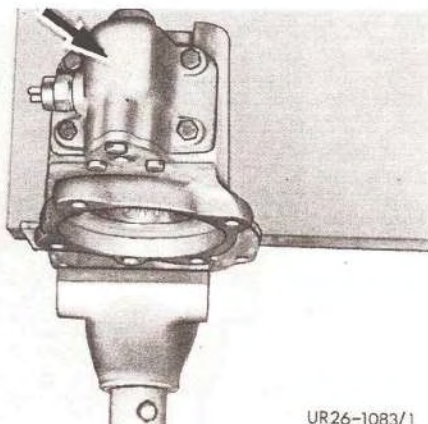


UR 26-1087/1

Assembly

- 1 Install deep-groove ball bearing and secure.
- 2 Install output shaft and secure.
- 3 Coat shaft sealing ring with grease No. 3 and install.





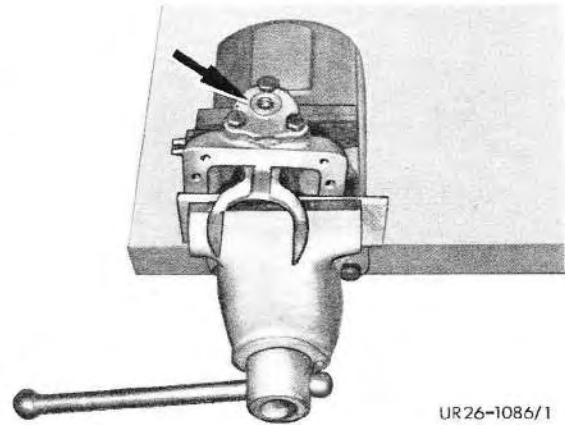
UR26-1083/1

- 4 Assemble shift housing, refer to 4.1/1.
- 5 Coat parting surface on shift housing with sealing compound No. 1 and bolts with sealing compound No. 2, fit shift housing and electric switch. Refer to 1.3/1 for perm. tightening torques.
- 6 Fit fast PTO, refer to 2.1/1.

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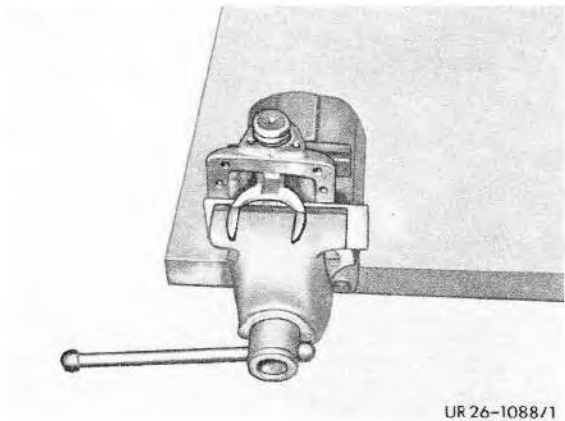
Disassembly

- 1 Remove shift housing, refer to 3.1/1.
- 2 Clamp shift housing in vice and remove bearing cover.



UR 26-1086/1

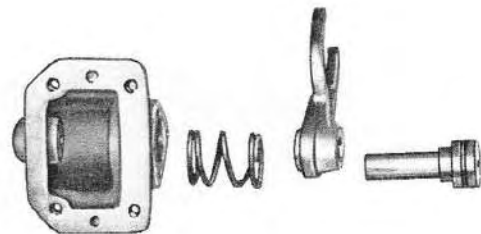
- 3 Remove shift piston, shift fork and compression spring.
- 4 Clean and check all parts, exchange as necessary.



UR 26-1088/1

Assembly

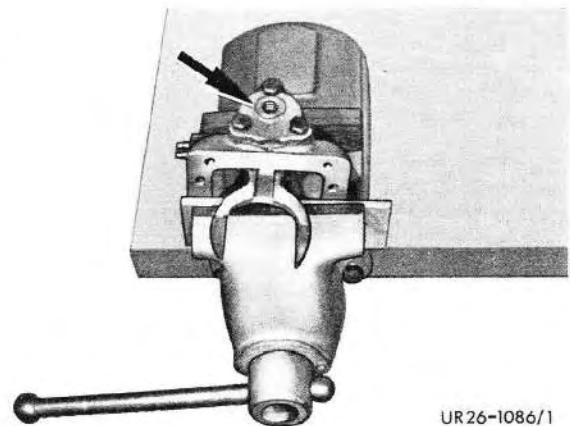
- 1 Install compression spring, shift fork and shift piston.



UR 26-1085

- 2 Fit bearing cover with gasket, coat bolts with sealing compound No. 2 and screw in. Remove shift housing from vice. Refer to 1.3/1 for perm. tightening torques.

- 3 Fit shift housing, refer to 3.1/2.



UR 26-1086/1

Contents

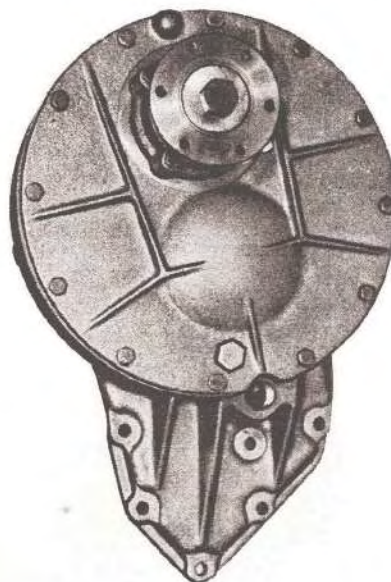
Chapter	Page
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2 Removing and installing power take-off	2.1/1
3 Disassembling and assembling power take-off	3.1/1

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Installation survey

Chassis		Main transmission		PTO Ratio i	Dir. of rotation	Installation SA
Model	Sales designation	Model	Sales designation			
435.110	U 1700	717.901	UG 3/40-8/13.01GPA	0.71	clockwise	35 925
.111	U 1700 L					
.115	U 1300					
.117	U 1300 L/37					

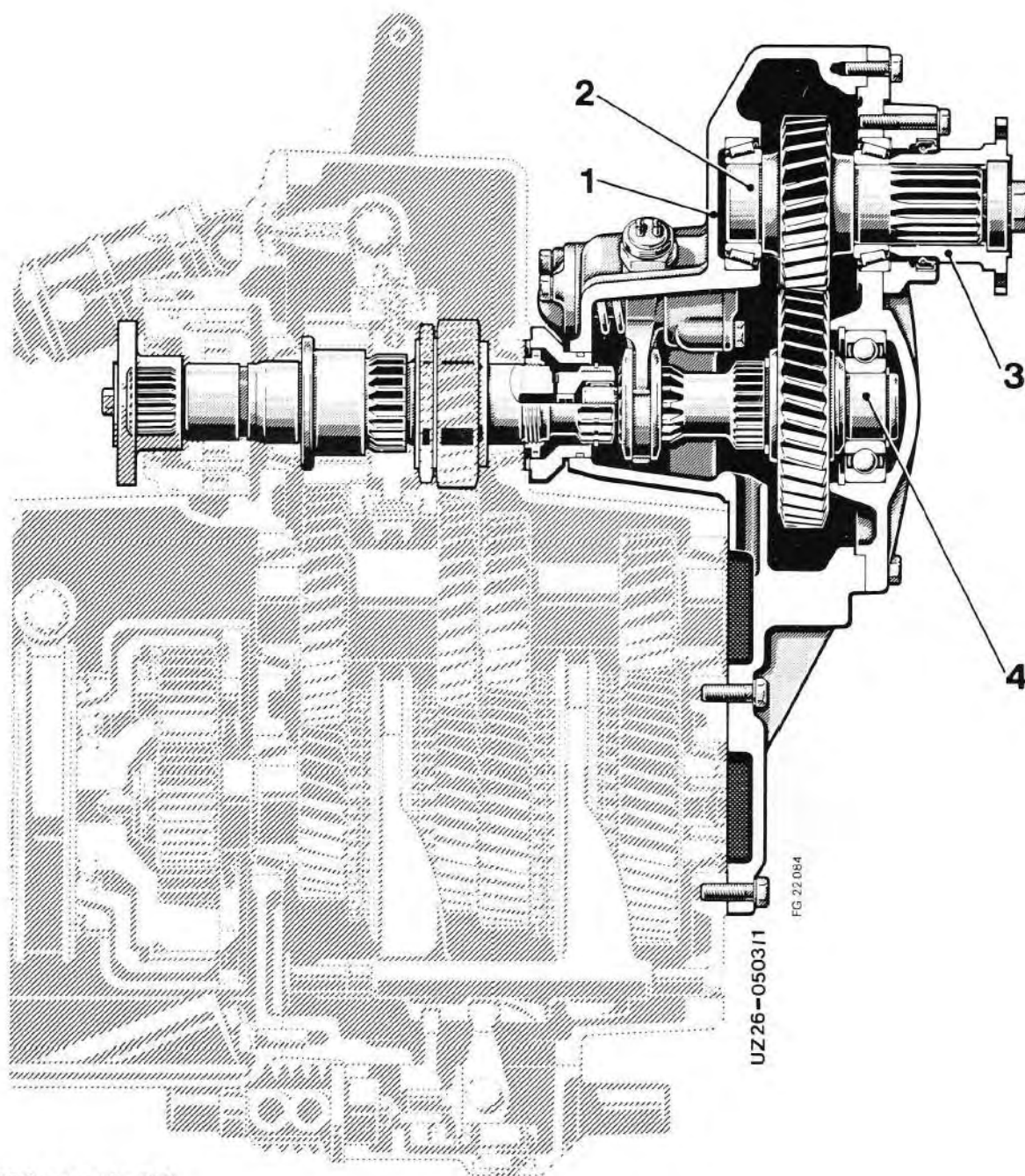
General View



UR26-1175

PTO
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Sectional View



PTO output $i = 0,71$

- 1 PTO output housing
- 2 Implement shaft
- 3 Implement flange
- 4 Input shaft

Capacities

Designation	Service product	SAE-grade	Capacity L	Installation
Main transmission	Transmission oil ¹⁾	80 ²⁾	11,0	Series
- with PTO		80 W/85 W ²⁾ 85 W-90 ³⁾ 90 ³⁾		SA 35 925

1) For U 435.111/.113/.115 special vehicles engine oil SAE 10 W

2) Optional

3) In hot climates

Settings

Axial play main shaft	mm	0,03 ± 0,02
Axial play output shaft	mm	0,03 to 0,06

Necessary Materials

Cons. No.	Designation	Part No.
1	Loctite No. 573 ¹⁾	001 989 45 20
2	Loctite No. 241 ²⁾	002 989 70 71
3	Longlife grease	comercially available

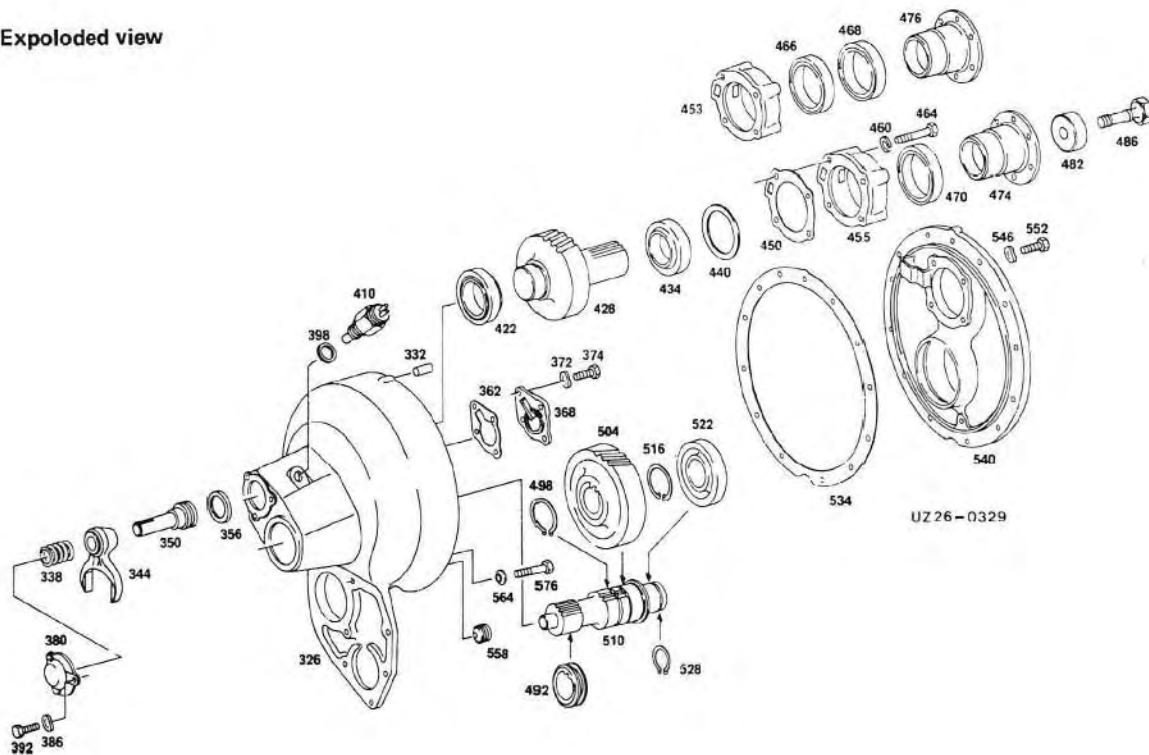
1) Surface sealing compound

2) Bolt sealing compound

Tightening Torques

Location	Thread-strength	Nm
Output flange mount	M 16 x 1,5 x 55 -10,9	170
PTO to main transmission	M 10 x 30 - 8.8	33
	M 10 x 105 - 8.8	33
Bearing cover to PTO	M 8 x 25 - 8.8	25
Bearing flange to bearing cover	M 8 x 45 - 8.8	25
Bearing cover to shift piston	M 8 x 18 - 8.8	25

Exploded view

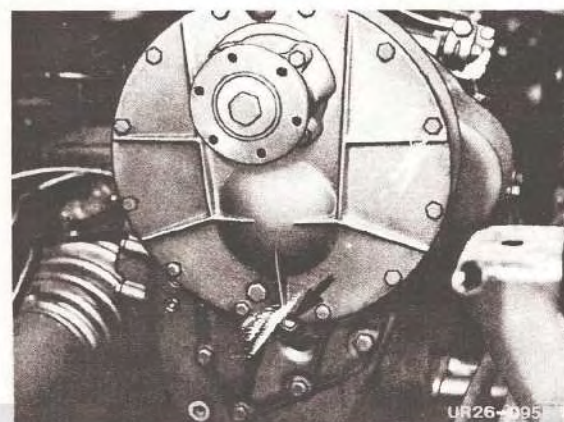
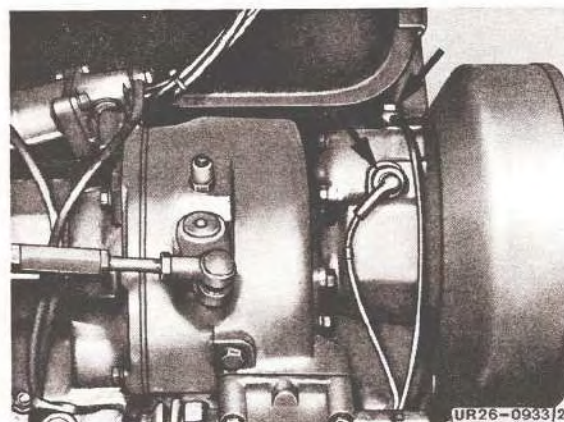


PTO

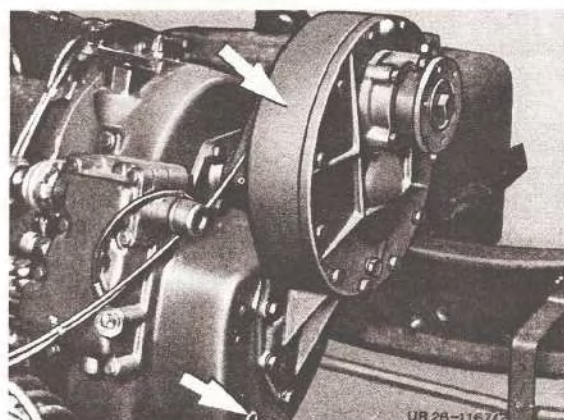
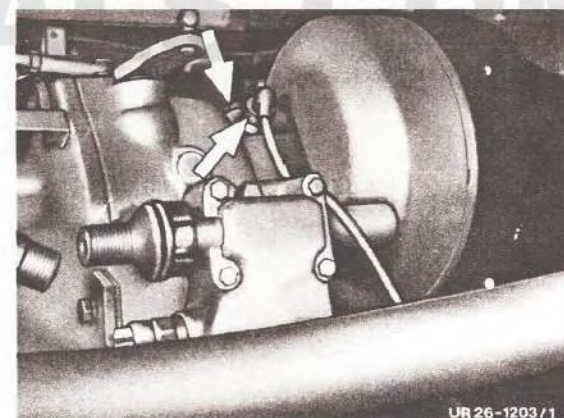
- | | | | |
|-----|--------------------|-----|--------------------|
| 326 | PTO housing | 464 | Bolt |
| 332 | Locking pin | 466 | Bearing |
| 338 | Compression spring | 468 | Bearing |
| 344 | Shift fork | 470 | Bearing |
| 350 | Shift cylinder | 474 | Output flange |
| 356 | Sealing ring | 476 | Output flange |
| 362 | Gasket | 482 | Supporting disk |
| 368 | Bearing cover | 486 | Bolt |
| 372 | Spring washer | 492 | Shift slide sleeve |
| 374 | Bolt | 498 | Spring washer |
| 380 | Bearing cover | 504 | Drive gear |
| 396 | Spring washer | 510 | Drive shaft |
| 392 | Bolt | 516 | Spring washer |
| 398 | Sealing ring | 522 | Bearing |
| 410 | Electr. switch | 528 | Spring washer |
| 422 | Bearing | 534 | Gasket |
| 428 | Output shaft | 540 | Bearing cover |
| 434 | Bearing | 546 | Spring washer |
| 440 | Spacer ring | 552 | Bolt |
| 450 | Gasket | 558 | Oil drain plug |
| 453 | Bearing flange | 564 | Spring washer |
| 455 | Bearing flange | 576 | Bolt |
| 460 | Spring washer | | |

Removal

- 1 Detach PTO shaft.
- 2 Unscrew pneumatic line and cable for indicator.
- 3 Drain oil, place support beneath power take-off and remove.

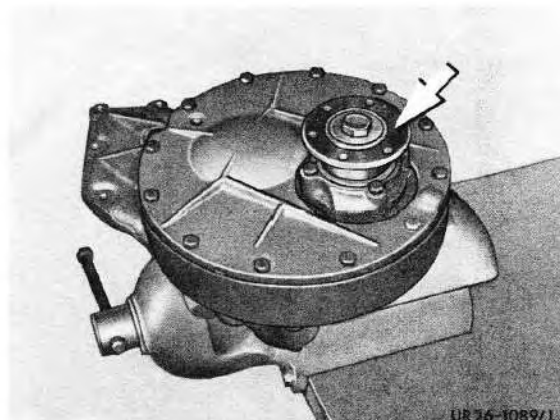
**Installation**

- 1 Coat contact surface with sealing compound No. 1 and attach power take-off.
- 2 Screw on pneumatic line and cable for indicator.
- 3 Fill with oil.
For filling capacity, see 1.4/1.
- 4 Attach PTO shaft.

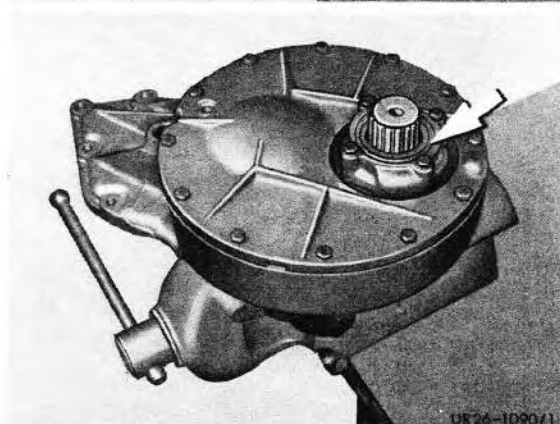


Disassembly

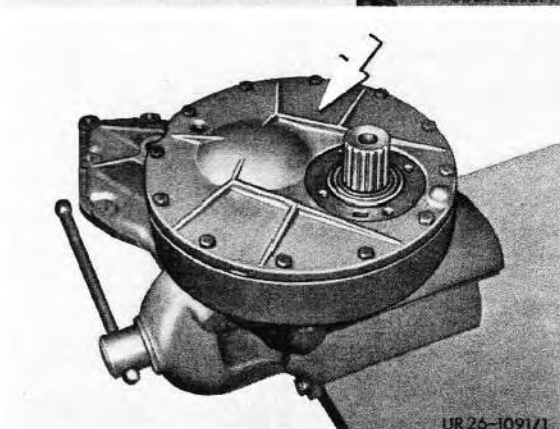
- 1 Remove fast PTO, refer to 2.1/1.
- 2 Clamp PTO in vice and remove output flange.



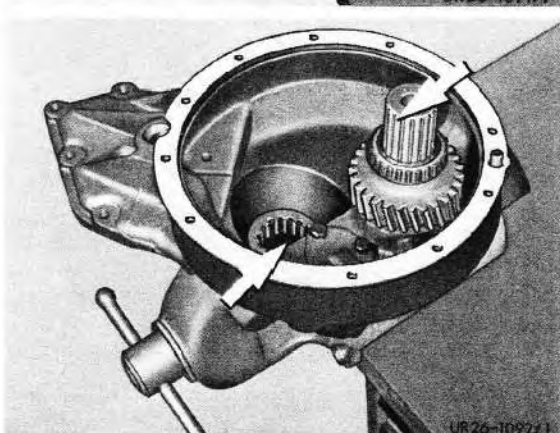
- 3 Remove bearing flange.

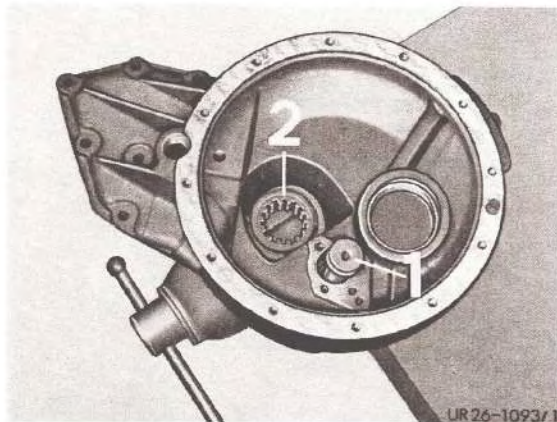


- 4 Remove bearing cover.

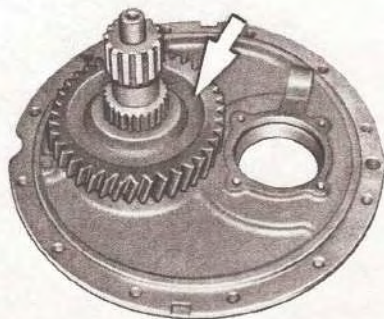


- 5 Remove output shaft and bearing cover.





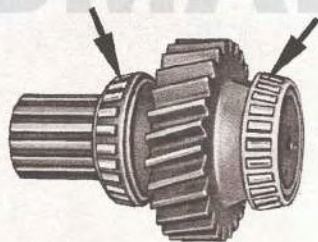
- 6 Remove shift piston (1), shift slide sleeve (2) and shift fork with compression spring.



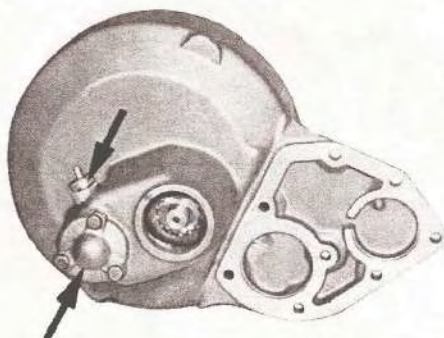
- 7 Release drive gear and remove.

- 8 Release deep-groove ball bearing in bearing cover and force out together with drive shaft.

- 9 Release deep-groove ball bearing on drive shaft and force out.



- 10 Pull off deep-groove ball bearing on output shaft.

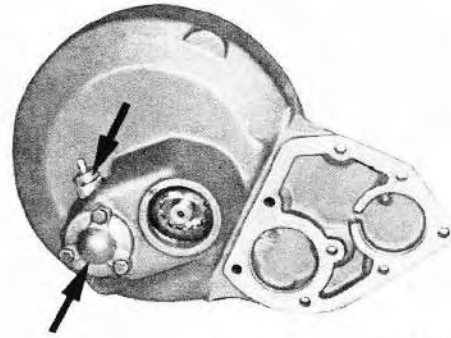


- 11 Remove PTO from vice and fit electric switch together with sealing and rear bearing cover.

- 12 Clean and check all parts, exchange as necessary.

Assembly

1 Coat rear bearing cover collar with sealing compound No. and fit. Fit electric switch with sealing ring.

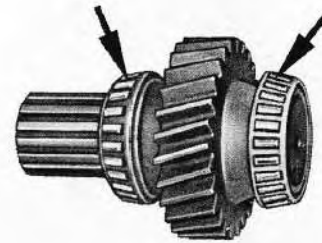


UR26-1102/1

2 Press in deep-groove ball bearing on output shaft.

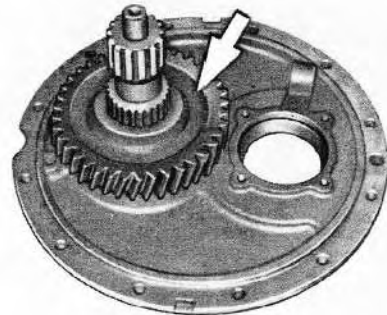
3 Press in deep-groove ball bearing on drive shaft and secure.

4 Press drive shaft with deep-groove ball bearing into bearing cover and secure.



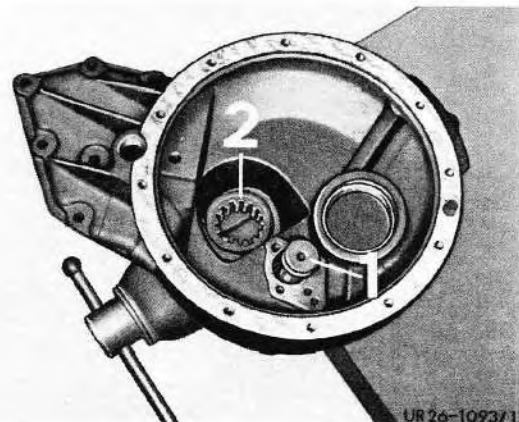
UR26-1095/1

5 Heat drive gear to approx. 160° , install and secure.

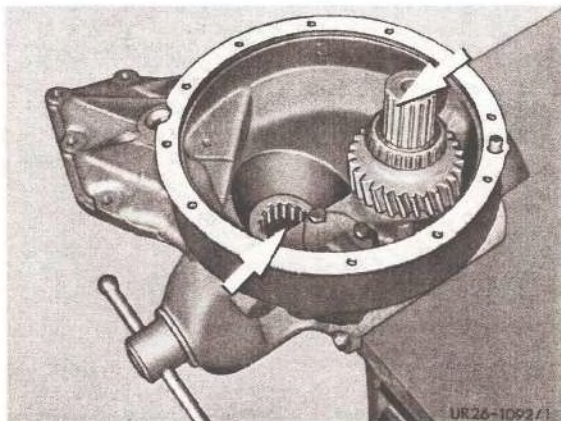


UR26-1094/1

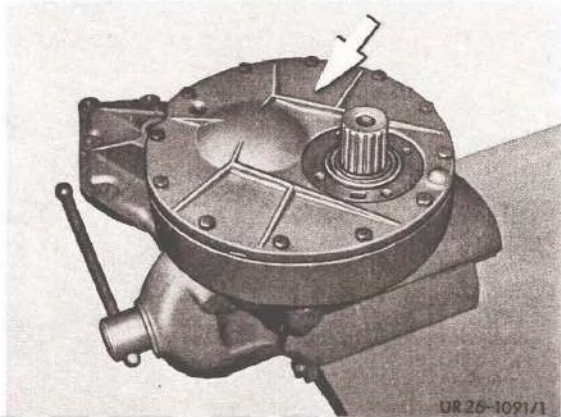
6 Install shift fork with compression spring, shift slide sleeve (2) and shift piston (1) with sealing ring.



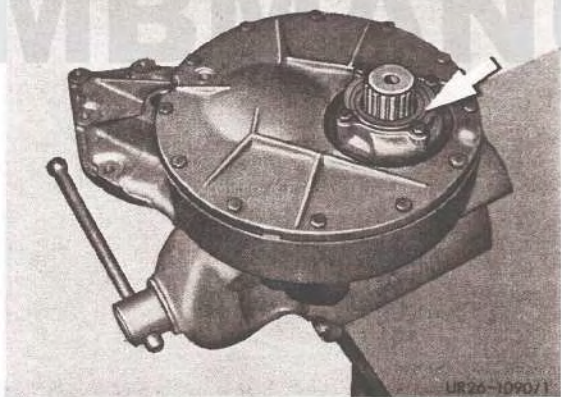
UR26-1093/1



- 7 Install output shaft and bearing cover.



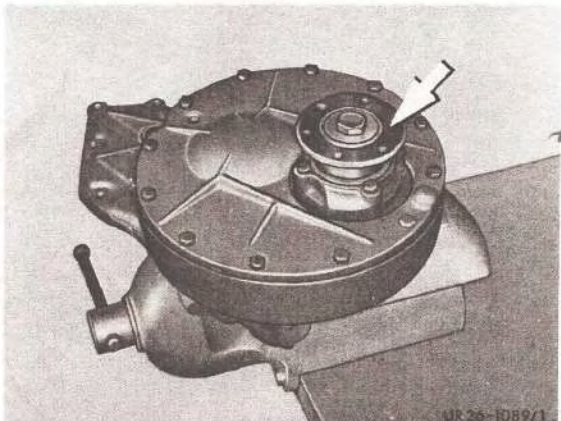
- 8 Fit bearing cover.



- 9 Measure bearing flange and fit. Refer to 1.4/1 for perm. end play.

- 10 Coat shaft sealing ring with grease No. 3 and install.

Note: Two shaft sealing rings are fitted with fording capability.



- 11 Fit output flange and remove PTO from vice. Refer to 1.4/1 for perm. tightening torques.

- 12 Fit fast PTO, refer to 2.1/1.

Survey

Version

Clutch Pedal

29.10

Brake Pedal

29.11

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Removal and installation of clutch master cylinder	3.1/1
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4 Ventilating clutch actuating	4.1/1
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6 Adjusting starter interlock	6.1/1

Installation Survey

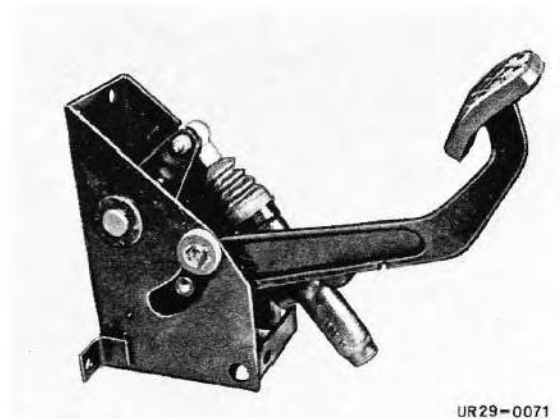
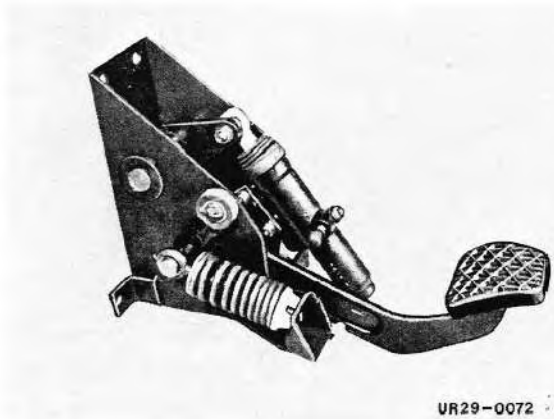
Chassis Model	Sales Designation	Series Chassis end No.	Cab Model	Clutch pedal Version	Installation
435.110	U 1700	from 002 230	425.800 425.820 425.821 425.825	With over-center helper spring	Series from PB ³⁾
.111	U 1700 L ¹⁾	from 000 008			
.113	U 1700 L/38	from 084 642			
.115	U 1300 L ¹⁾	from 000 019		Without over-center helper spring ²⁾	SA 35 893 ²⁾ from PB ³⁾
.117	U 1300 L/37	from 071 796			
.170	Attachment Condor	from PB ³⁾	—	With over-center helper spring	Series from PB ³⁾

1) L = Truck

2) Over-center helper spring not installed in conjunction with PTO transmission (SA 35 737)

3) PB = Start of production

General View



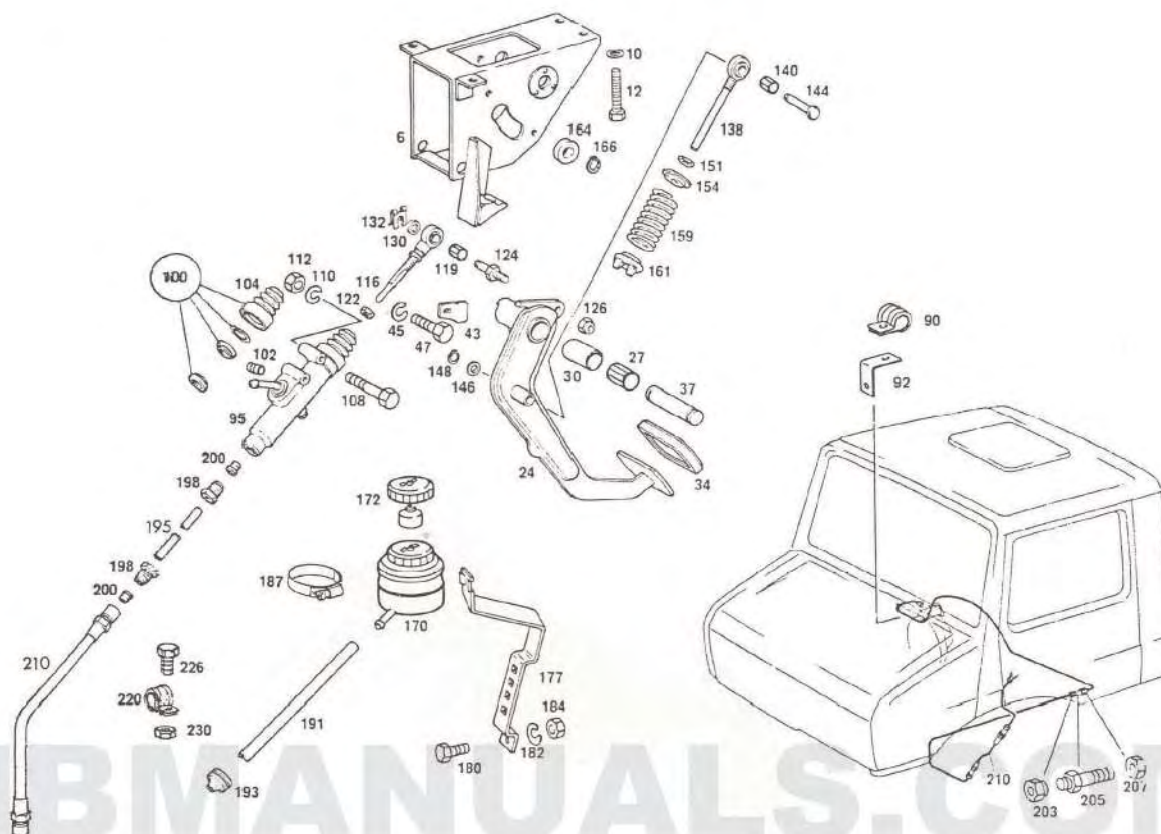
Clutch pedal

Clutch pedal (SA 35 893)

Special tools

Cons. No.	Tool	Special tool	Tool set
1	Connection piece for basic setting of clutch pedal	425 589 00 90 00	B
2	Tensioner for over-center helper spring	425 589 03 31 00	B

Exploded view



U Z 29-0021

Clutch pedal gear

6 Bearing bracket	112 Nut	170 Reservoir
10 Spring washer	116 Rod	172 Cap
12 Bolt	119 Needle bearing	177 Bracket
24 Pedal	122 Nut	180 Bolt
27 Needle bearing	124 Pin	182 Lock washer
30 Tube	126 Nut	184 Nut
34 Cap	130 Washer	187 Expanding clip
37 Pivot	132 Retainer	191 Hose
43 Retainer	138 Rod	193 Grommet
45 Lock washer	140 Needle bearing	195 Line
47 Bolt	144 Pin	198 Nut
90 Clamp	146 Washer	200 Bite ring
92 Bracket	148 Snap ring	203 Cap nut
95 Master cylinder	151 Nut	205 Bulkhead lead-in
100 Repair set	154 Spring seat	207 Nut
102 Dust cap	159 Spring	210 Hose
104 Boot	161 Spring seat	220 Clamp
108 Bolt	164 Stop	226 Bolt
110 Lock washer	166 Snap ring	230 Nut

Removal**CAUTION: DANGER**

If mounted top dead center spring, watch for sudden dropping of clutch pedal.

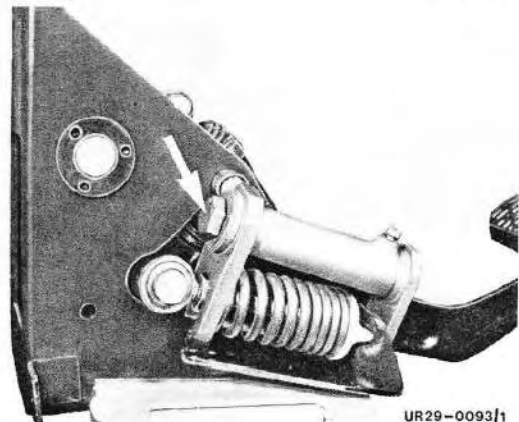
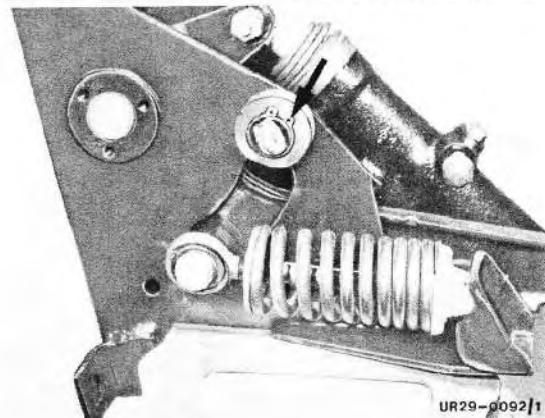
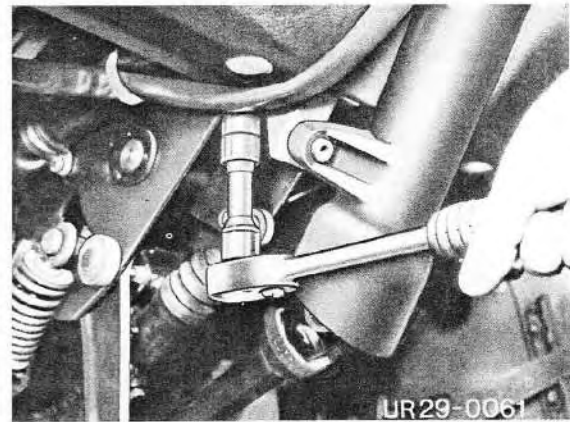
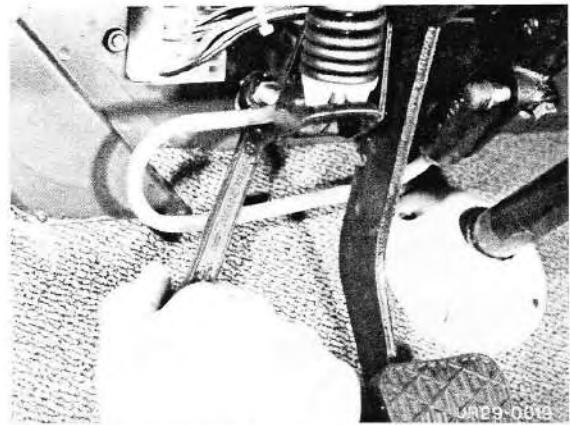
- 1 Detach line at clutch master cylinder.
- 2 Unscrew pipeline at connector.
- 3 Detach wiring harness clamp from pedal bracket.
- 4 Unscrew and remove pedal bracket.

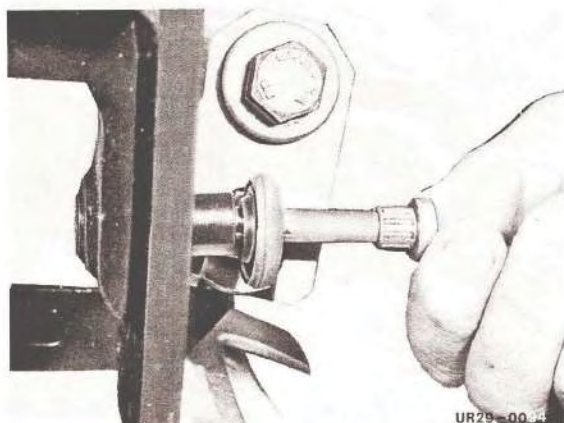
Caution:

Danger — watch out for sudden dropping of clutch pedal.

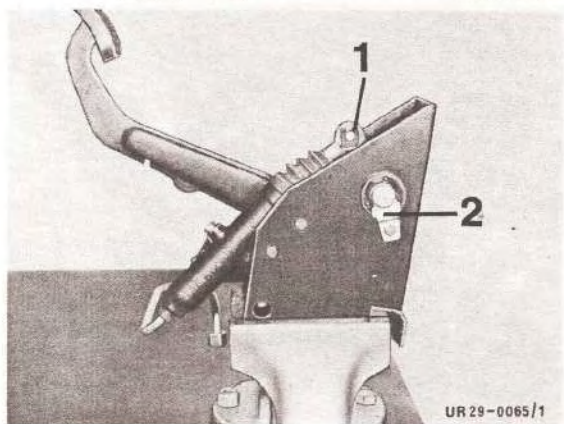
- 5 Clamp clutch pedal and detach stop damper.

- 6 Release over-center helper spring on push rod, attach special tool No. 2 and compress over-center helper spring.

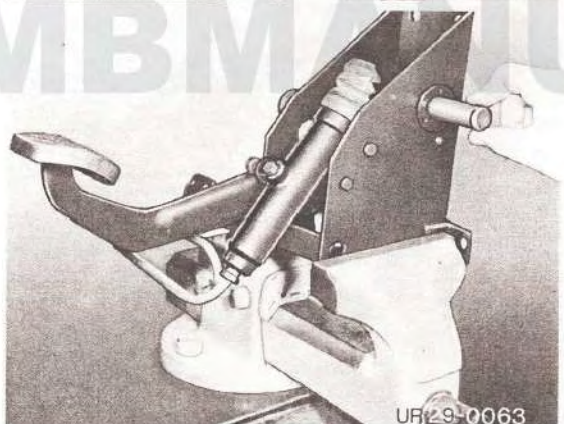




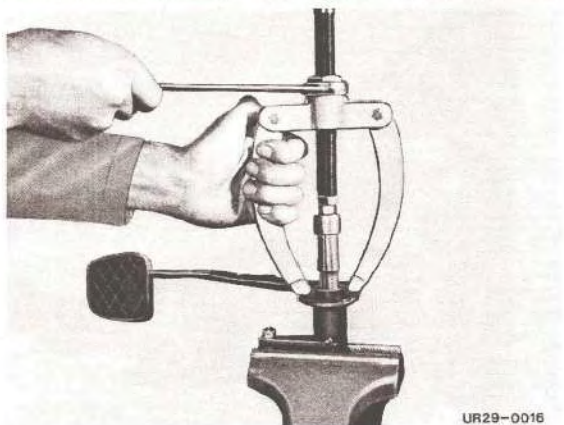
- 7 Release and withdraw spring retaining pin.
Using special tool No. 2, remove push rod with needle bearing, as well as spring and spring seat.



- 8 Force clutch pedal upwards.
9 Release push rod (1) of clutch master cylinder and unscrew retainer at bearing pin (2).



- 10 Withdraw bearing pin.



- 11 Draw both needle bearings out of clutch pedal.
12 Clean and check all parts, exchanging if necessary.